GEOGRAPHY

PHYSICAL GEOGRAPHY





MODULE - 12

- INTERIOR OF THE EARTH
- DIRECT AND INDIRECT SOURCES
- GEOTHERMAL GRADIENT
- CHANGE OF GRAVITATION

VOLCANIC ERUPTION

EARTHQUAKE

TSUNAMI

SURFACE OF THE EARTH

HIMALAYAS

GREAT RIFT VALLEY, AFRICA

GREAT LAKES REGION

Earth's radius is 6,370 km. How do one see inside it?

INTERIOR OF THE EARTH

DIRECT SOURCES

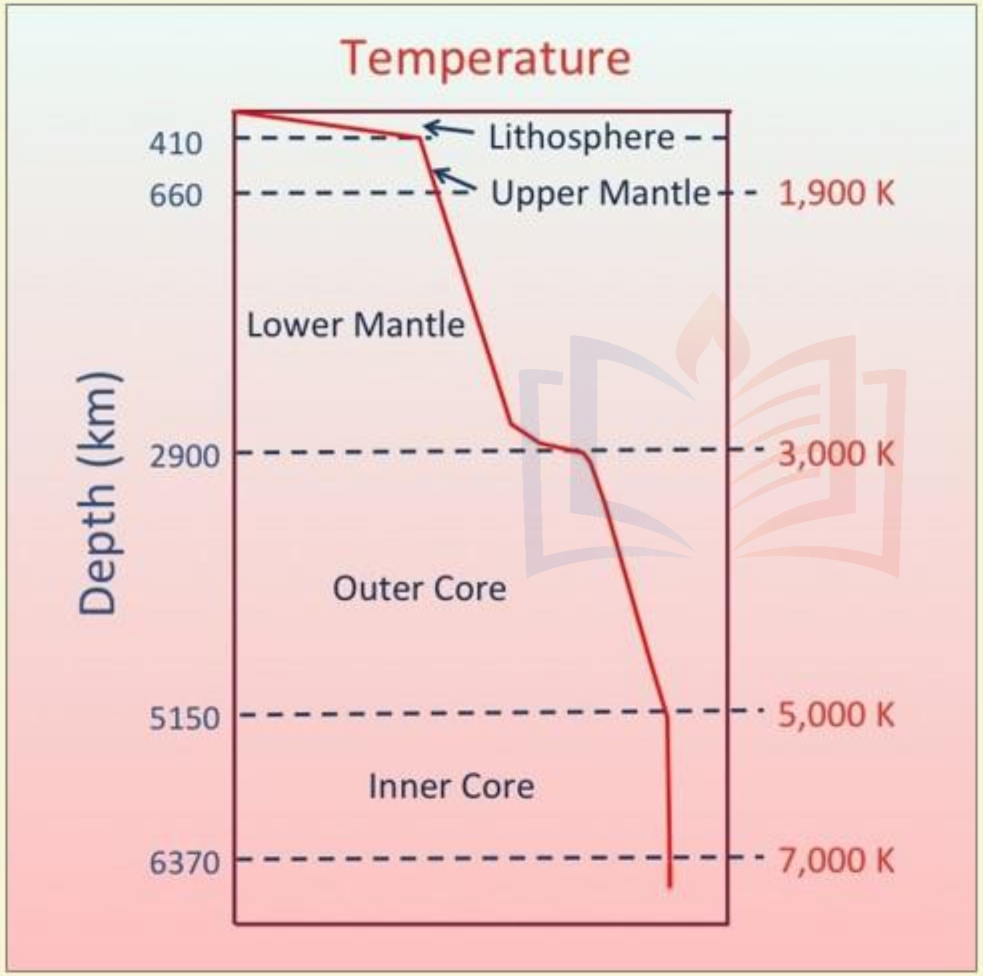
- Surface rocks.
- Rocks from the mining areas.
- Information from Deep Sea Drilling Project.
- Information from Integrated Ocean Drilling Project.
- Other Deep Drilling Projects.
- Volcanic eruptions.

HOW FAR CAN WE GO?

- We can go up to only a few kilometres, say 3 to 4 km.
- It is very hot even at this depth.

INTERIOR OF THE EARTH

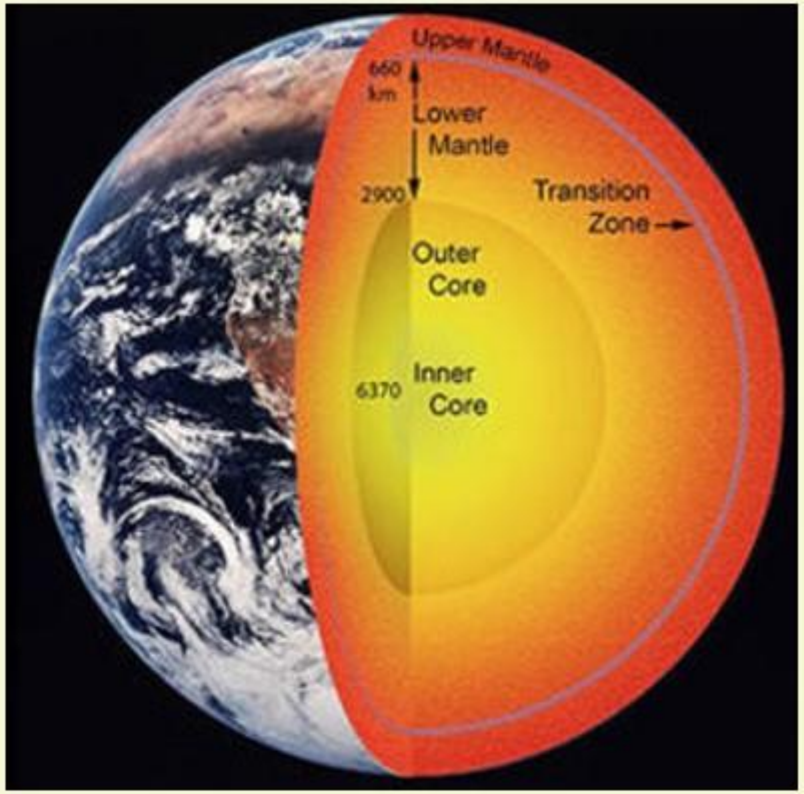
DIRECT SOURCES



GEOTHERMAL GRADIENT

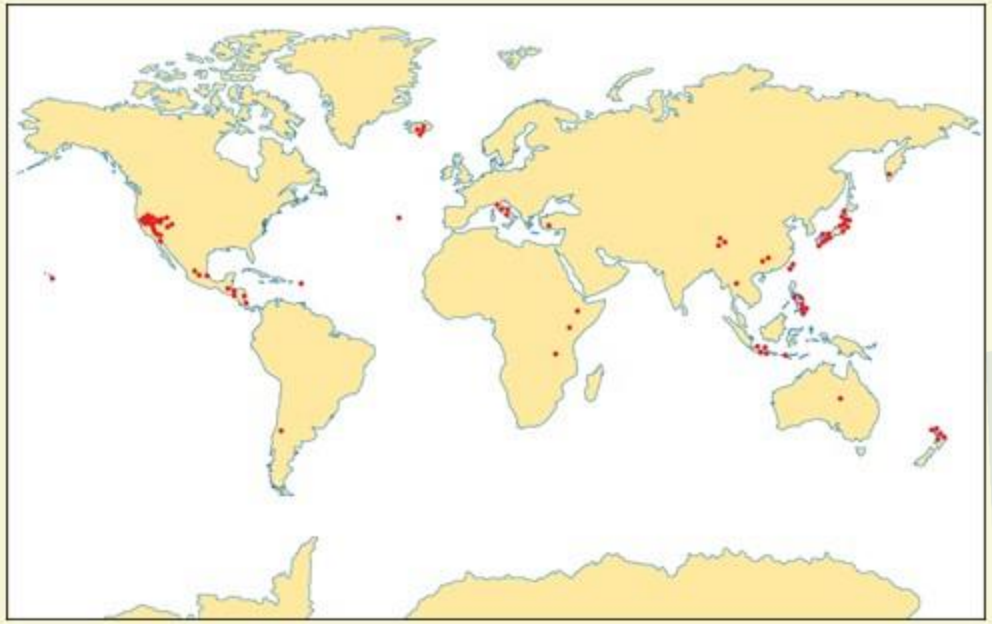
1. It is the amount the Earth's temperature increases with depth.
2. It indicates heat flowing from the Earth's warm interior to its surface.
3. On an average, the temperature increases by about 25-30°C per every kilometre of depth from the surface of the Earth.
4. The difference in temperature drives the geothermal energy flows and it allows humans to use this energy for heating and electricity purposes.

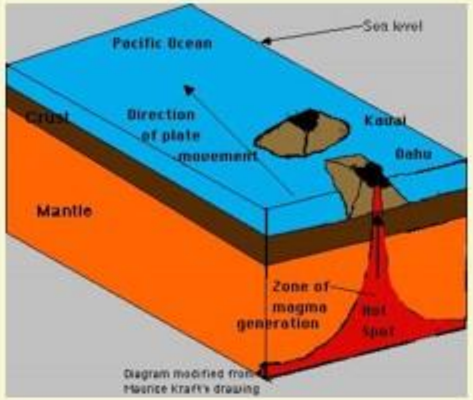
FROM WHERE DOES THIS HEAT COMES FROM?

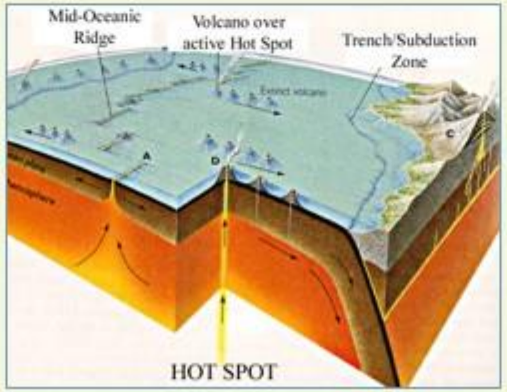


1. It is discovered in the early 20th Century that the Earth's underground heat is from radio-active elements.
2. The heating is caused by the decay of elements such as Uranium, Thorium.
3. This is said to account for 50% of the Earth's heat.
4. The remaining heat is coming from the primordial heat of the Earth, that means, the heat from the Earth's formation.

CAN WE USE THIS GEOTHERMAL ENERGY?

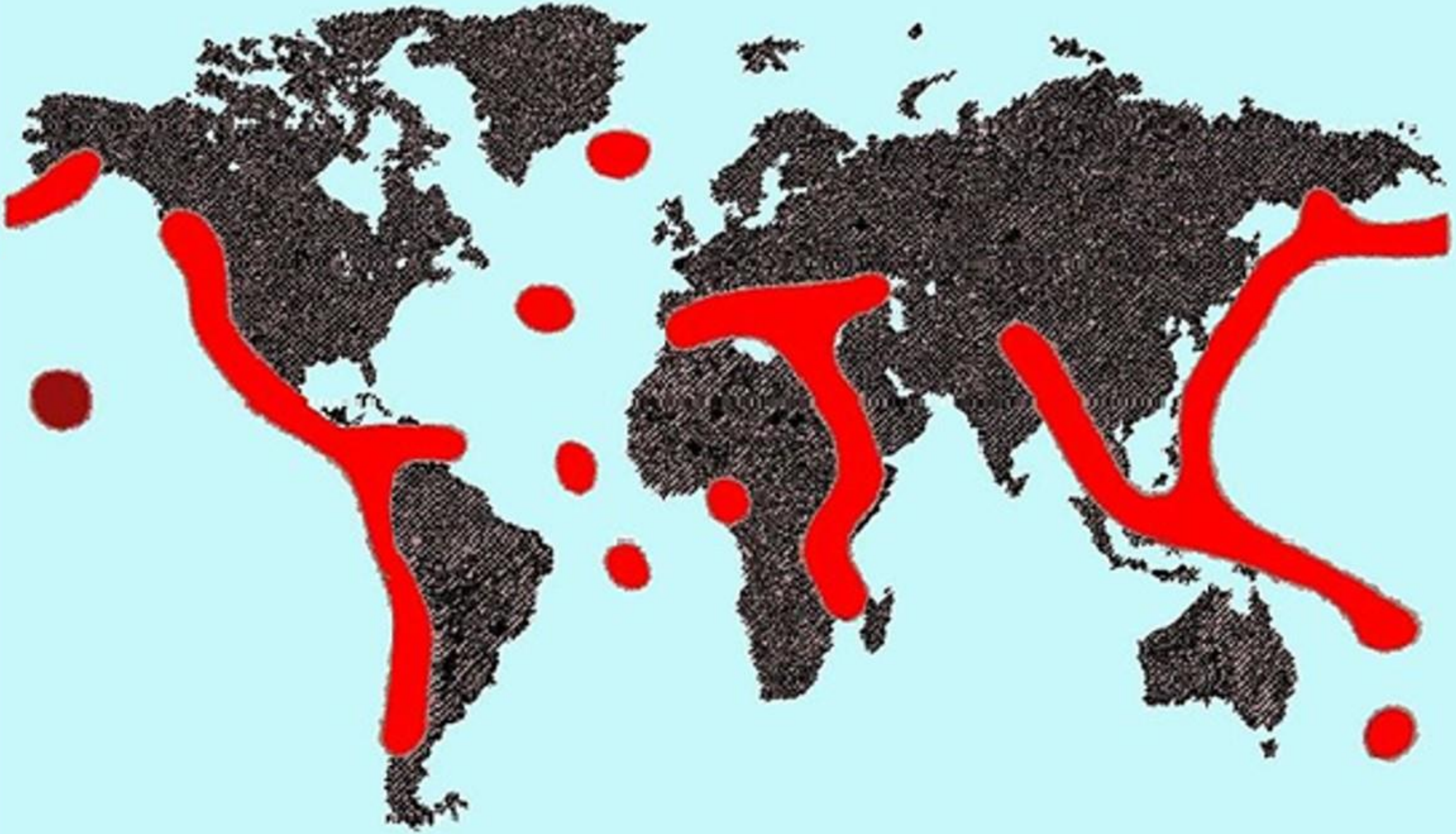






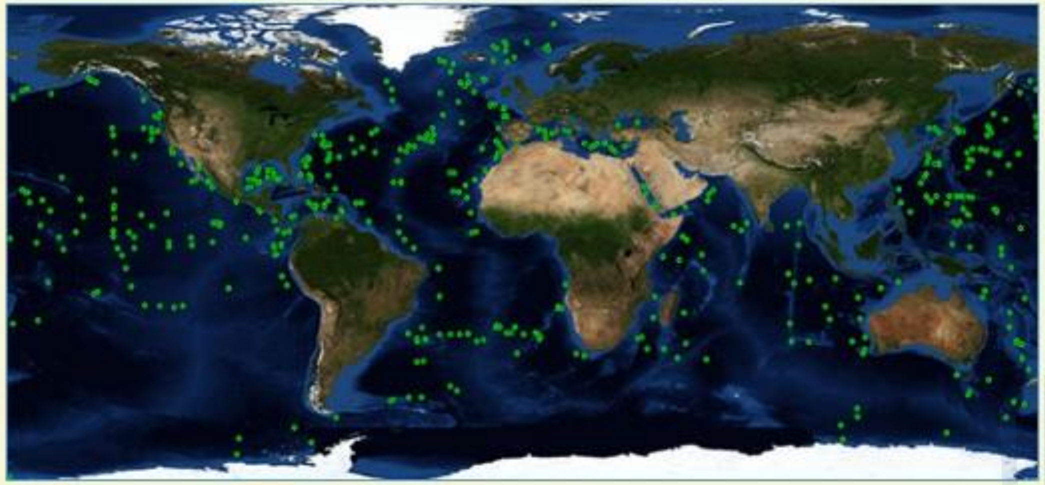
1. For most of the areas, the energy available for human use is low quality.
2. If the Country is situated on a hotspot, then we can tap that energy.
3. The hotspots occur, because of their location near Tectonic Plate Boundaries. Here the crust is thinner and hot magma extends close to the surface.

Geothermal hotspots



INTERIOR OF THE EARTH

DIRECT SOURCES



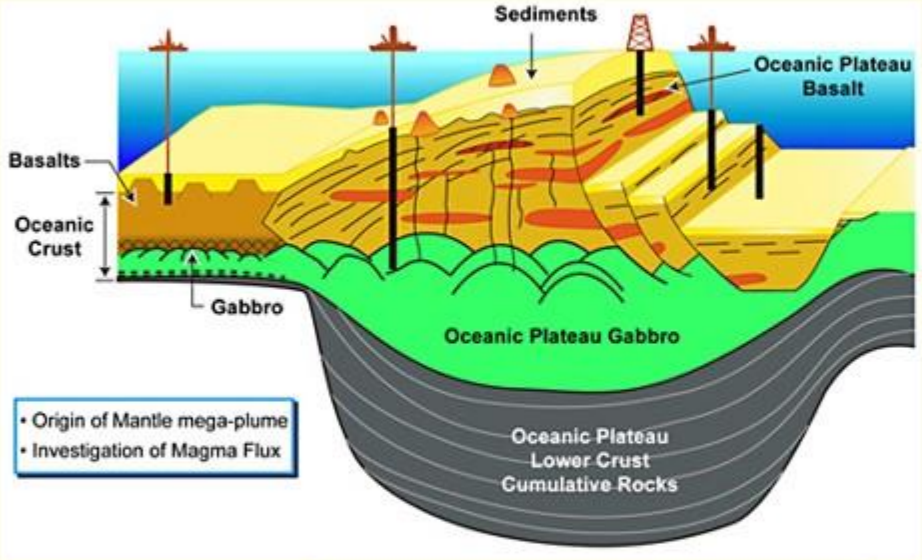
DEEP SEA DRILLING PROJECT

1. All began on June 24, 1966, when contract was signed between National Science Foundation and The Regents, University of California.
2. It is a scientific program for drilling cores of sediment and basaltic crust beneath the deep oceans.
3. The ship used was Glomar Challenger. It ended in 1983.
4. Though it was started primarily by National Science Foundation of USA, later on Countries like France, Britain, West-Germany, Japan and the USSR also participated.

The DSDP was succeeded in 1984 by the Ocean Drilling Program.

GLOMAR CHALLENGER

INTERIOR OF THE EARTH
DIRECT SOURCES



OCEAN DRILLING PROGRAM

1. Ocean Drilling Program is an international effort to explore and study the composition and structure of the Earth's ocean basins.
2. It began in 1985. Several countries participated.
3. The Program used the Drillship, JOIDES Resolution. JOIDES Resolution stands for Joint Oceanographic Institutions for Deep Earth Sampling.

In 2004, ODP transformed into Integrated Ocean Drilling Program.

JOIDES RESOLUTION

INTERIOR OF THE EARTH

DIRECT SOURCES

IODP
INTERNATIONAL OCEAN DISCOVERY PROGRAM

INTEGRATED OCEAN DRILLING PROGRAMME

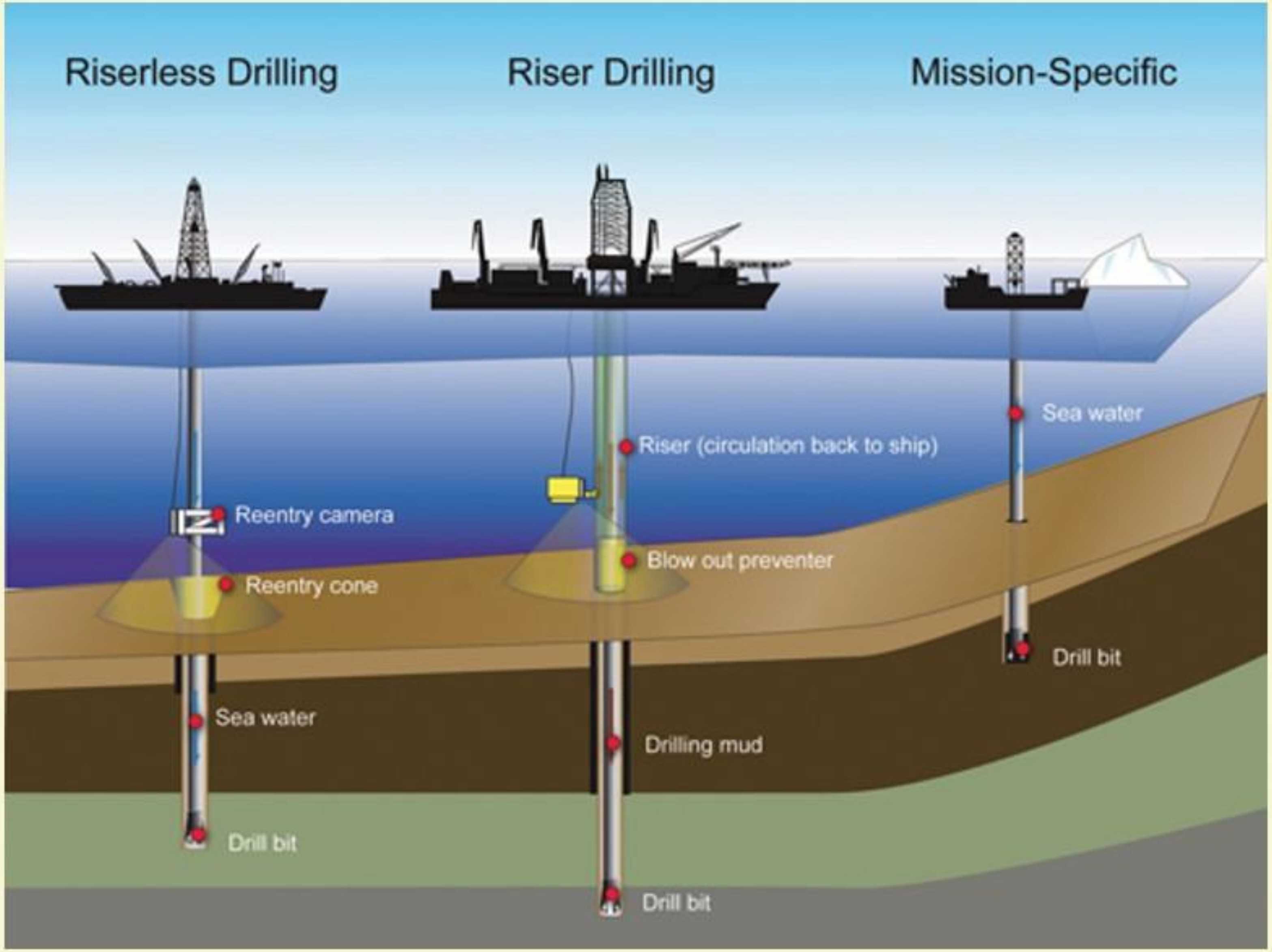
1. It is an International Marine Research Program.
2. It used heavy drilling equipment to monitor and sample sub-seafloor environments.
3. With this research, IODP documented environmental change, earth processes and effects, the biosphere, solid earth cycles, geodynamics.

It brought together Scientists from different Countries. National Consortia and Government Funding Agencies supported IODP.

MEXT
MINISTRY OF EDUCATION, CULTURE, SPORTS, SCIENCE AND TECHNOLOGY-JAPAN

NSF

ECORD
EUROPEAN CONSORTIUM FOR OCEAN RESEARCH DRILLING



INTERIOR OF THE EARTH

DIRECT SOURCES

सत्यमेव जयते
Ministry of Earth Sciences
Government of India

NATIONAL CENTRE FOR ANTARCTIC & OCEAN RESEARCH
MINISTRY OF EARTH SCIENCES (GOVT. OF INDIA)
NCAOR
- GOA -

WHAT ABOUT INDIA?

1. India joined the IODP, as an Associate Member during 2008-09.
2. A formal MoU was also signed between MoES and NSF / MEXT, the two lead agencies for IODP operations.
3. In India, the Participating Institute is National Centre for Antarctic and Ocean Research, Goa.

INTERIOR OF THE EARTH

INDIRECT SOURCES

1 BASED ON KNOWN PARAMETERS

Assessment of temperature, pressure and density based on the known parameters up to some distance.

2 METEORITE

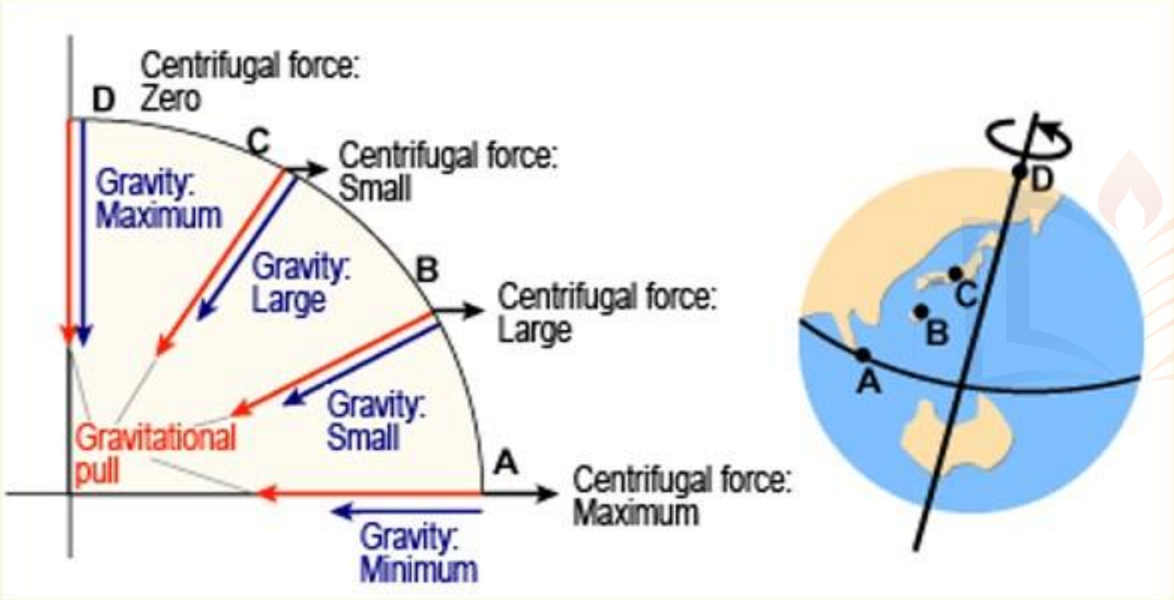
- Meteorites, because the material and the structure observed is similar to the Earth.
- These are solid bodies developed out of meteoroids same as or similar to our planet.

METEOROIDS, METEORS, METEORITES

- Meteoroid is a small particle from a comet or asteroid orbiting the Sun.
- When it enters the Earth's atmosphere and vaporizes, it results in a light phenomena, which is known as a Shooting Star or Meteor.
- A meteoroid that survives its passage through the Earth's atmosphere and which lands upon the Earth's surface is known as Meteorite.

INTERIOR OF THE EARTH

INDIRECT SOURCES



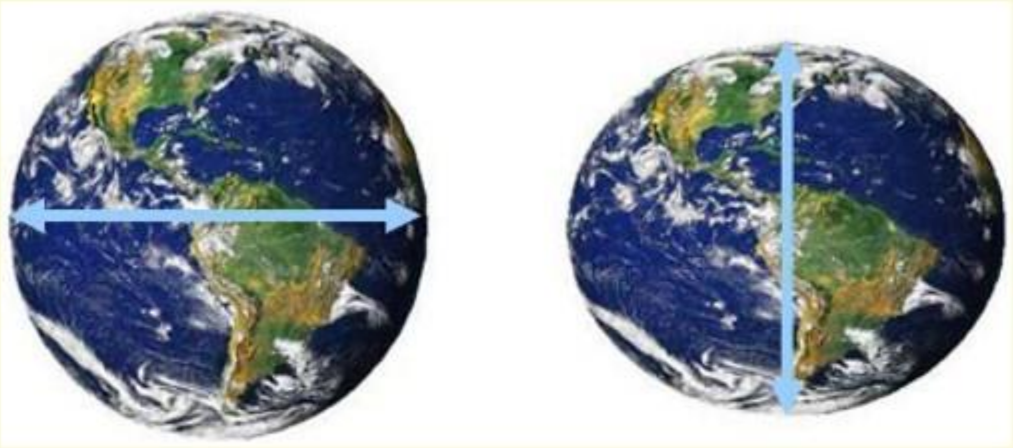
3 GRAVITATION

- Gravitation force is not the same at different latitudes.
- It is greater near the Poles and less at the Equator.
- This is because the distance from the Centre of the Earth is greater to the Equator than to the Poles.
- Gravity changes based on the mass of the materials.

Uneven distribution of the mass of the material within the Earth influences this value. Hence the reading of gravitation at different places is influenced by several other factors.

INTERIOR OF THE EARTH

INDIRECT SOURCES

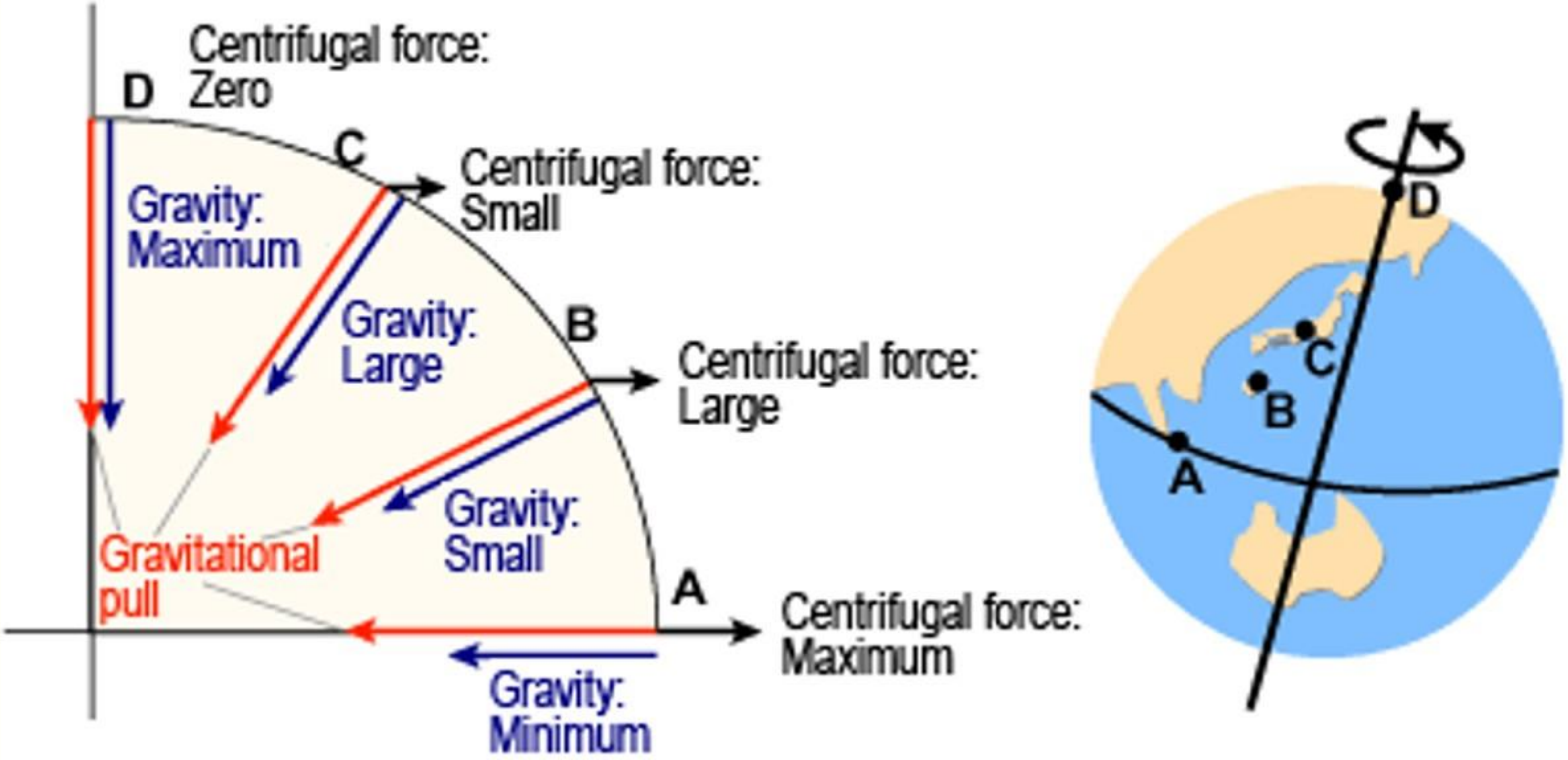


- *Radius of the Earth at*
  Poles .... 6,356 Km
- *Radius of Earth at*
  Equator .... 6,378 Km
- *Small differences due to local geology.*

3 GRAVITATION

WHAT ARE THE FACTORS THAT INFLUENCE CHANGE IN GRAVITATION?

- It is greater at the Poles than Equator, because of distance from Centre.
- Because of centrifugal force, gravitation is less at Equator.
- Gravitation is more at Dead Sea in comparison to Mount Everest.


Centrifugal force:
Zero
D
C
Centrifugal force:
Small
Gravity:
Maximum
Gravity:
Large
B
Centrifugal force:
Large
Gravity:
Small
Gravitational
pull
A
Centrifugal force:
Maximum
Gravity:
Minimum
D
C
B
A

DEAD SEA
- 430 m
MOUNT EVEREST
+ 8,848 m
Black Sea
Caspian Sea
TURKEY
GEORGIA
ARMENIA
AZERBAIJAN
UZBEKISTAN
KYRGYZSTAN
TURKMENISTAN
TAJIKISTAN
SYRIA
IRAQ
IRAN
AFGHANISTAN
PAKISTAN
CHINA
NEPAL
INDIA
JORDAN
EGYPT
SAUDI ARABIA
Persian Gulf
Gulf of Oman
OMAN
YEMEN
SUDAN
ERITREA
Red Sea
Arabian Sea
Bay of Bengal
MYANMAR
(BURMA)
VIETNAM
Lao PDR
THAILAND
CAMBODIA
Gulf of Thailand
South China Sea
ETHIOPIA
SOUTH SUDAN
SOMALIA
UGANDA
KENYA
TANZANIA

INTERIOR OF THE EARTH

INDIRECT SOURCES

4 MAGNETIC SURVEYS

Distribution of magnetic materials in the crust and information about the distribution of materials in crust.

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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To Contact us:
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OUR TEAM

G. V. Rao

Amrita Naidu	K. Srikanth
D. Sunil Kumar	G. Ravi Babu
L. V. Krishna	D. Joji
D. Chaitanya	K. Victor Babu